

# Answers for Solar Installation Technology Assessment

## SECTION A: SHORT RESPONSE QUESTIONS

1. State three main types of photovoltaic cells:

- Monocrystalline silicon cells
- Polycrystalline silicon cells
- Thin-film photovoltaic cells

2. Identify any four components of solar photovoltaic cells:

- Solar panel
- Inverter
- Charge controller
- Battery

3. To be able to calculate the size of a solar panel, there are a number of factors to be considered.

Name any four:

- Energy consumption needs
- Available sunlight (irradiance)
- Efficiency of the solar panel
- Space available for installation

4. Name any four factors that affect solar panel efficiency:

- Temperature
- Dirt, dust, or shading
- Quality of materials used in the panel

- Orientation and tilt angle

5. State two main methods of solar panel connection:

- Series connection
- Parallel connection

6. State two types of cables:

- DC cables (e.g., PV cables)
- AC cables

7. Identify three types of mounting systems:

- Roof-mounted systems
- Ground-mounted systems
- Pole-mounted systems

8. State three factors to consider during solar panel installation:

- Location and sunlight exposure
- Structural integrity of the mounting surface
- Orientation and tilt of the solar panels

9. Laying cables in a solar system installation requires a set of tools designed to handle various aspects of cable management and installation. State any five cable laying tools:

- Wire strippers
- Crimping tools
- Cable cutters
- Conduit benders
- Cable testers

10. Identify any three Personal Protective Equipment (PPE):

- Safety gloves
- Safety goggles
- Hard hat

11. Identify any six key aspects involved in cable hanging:

- Proper tensioning of cables
- Avoiding sharp bends or kinks
- Using appropriate cable trays or conduits
- Securing cables with proper fasteners
- Maintaining sufficient clearance from heat sources
- Labeling and organizing cables properly

## SECTION B: EXTENDED RESPONSE QUESTIONS

12(a) State and explain any three roles of a charge controller:

- Regulating voltage and current: Prevents overcharging of batteries.
- Battery protection: Ensures batteries are not over-discharged.
- Power management: Distributes power effectively between the load and battery.

12(b) With an aid of a diagram, explain three methods of solar panel connection:

- Series connection: Voltage increases while current remains constant.
- Parallel connection: Current increases while voltage remains constant.
- Mixed connection: Combination of series and parallel for balanced output.

13(a) State and explain four types of charge controllers:

- PWM (Pulse Width Modulation): Simple and cost-effective, but less efficient.
- MPPT (Maximum Power Point Tracking): Optimizes power transfer, highly efficient.
- On/Off controllers: Simple control by switching charging on or off.
- Hybrid controllers: Combines features of PWM and MPPT for versatile usage.

13(b) Discuss six key aspects of cable segregation:

- Segregating power and data cables to avoid interference
- Labeling cables for easy identification
- Bundling cables based on function
- Providing adequate spacing between high-voltage and low-voltage cables
- Using color coding for different types of cables
- Routing cables through separate conduits or trays

14(a) Discuss five factors to consider in terminating electrical cables:

- Using the correct termination tools
- Ensuring proper insulation stripping without damaging conductors
- Choosing appropriate connectors for the cable type
- Ensuring secure and tight connections
- Avoiding moisture or contamination at termination points

14(b) Discuss five factors to consider for cable connectors used in insulation:

- Material compatibility with the cable insulation
- Current-carrying capacity of the connector
- Resistance to environmental conditions (e.g., UV, heat)
- Ease of installation and maintenance
- Durability and mechanical strength